

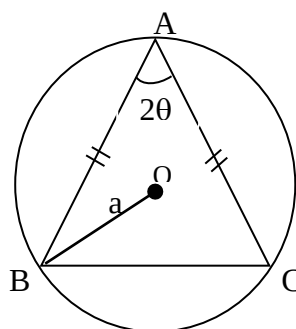
S.6 END OF MARCH TEST 2006

MATHS

TIME: 3 HOURS

1. (i) Show that  $\frac{1}{2}$  is a root of the equation  $2Z^3 - 9Z^2 + 14Z - 5 = 0$ . Hence find the other roots  
Express  $Z$  in polar form
- (ii) The points  $A, B$  and  $C$  represent the complex numbers  $Z_1, Z_2$  and  $Z_3$  respectively and  $ABC$  is an isosceles triangle with a right angle at  $A$ . Prove that  $2Z_1^2 + Z_2^2 + Z_3^2 = 2(Z_3Z_1 + Z_1Z_2)$
2. Find the equation of the chord joining the points  $(at_1^2, 2at_1)$  and  $(at_2^2, 2at_2)$  of the parabola  $y^2 = 4ax$ 
  - (i) Show that it cuts the directrix when  $y = \frac{2a(t_1t_2 - 1)}{(t_1 + t_2)}$
  - (ii) Deduce the equation of the tangent at  $(at^2, 2at)$  and prove that the tangents at the ends of a focal chord intersect at right angles on the directrix.

3. (a)



In the figure above,  $ABC$  is an isosceles triangle of vertical angle  $2\theta$  inscribed in a circle of radius  $a$ . Show that the area of the triangle is  $4a^2 \sin\theta \cos^2\theta$   
Hence determine the maximum area and the value of  $\theta$  for which it occurs.  
Comment on your answer.

(b) Prove that in any triangle  $ABC$ ,  $\sin^2A + \sin^2B - \sin^2C = 2\sin A \sin B \cos C$

(c) Find the ortho-centre of the triangle with the vertices at the points  $(3, -1), (-2, 3)$  and  $\left(-\frac{36}{7}, -\frac{45}{7}\right)$

4. If  $A, B$  are the points  $(3,0), (0,4)$  respectively, show that the locus of a point  $P$  such that  $4PA = 3PB$  is the circle  $7(x^2 + y^2) - 96x + 72y = 0$ .  
Find the equation of the tangent to this circle at right angles to the line  $(2x - 3)^2 + (2y - 4)^2 = 25$ .

5. (a) A party of nine people travel in two cars, one of which will not hold more than seven people and the other not more than four. In how many ways can the party travel?
- (b) There are ten candidates for three vacancies and an elector can vote for any number of candidates not greater than the number of vacancies. In how many ways can an elector vote?
- (c) In how many ways can three consonants and two vowels be chosen from the word LOGARITHMS and in how many of these will the letter i occur?
6. (a) solve  $\log(x-1) + 2\log y = 2\log 3$   
 $\log x + \log y = \log 6$
- (b) Find the pairs of x and y which satisfy the equations  
 $x^3 + y^3 = 35$   
 $x^2y + xy^2 = 30$
- (c) Solve
- (i)  $\sqrt{x^2 - 3x + 6} - \sqrt{x^2 - 3x + 3} = 1$
- (ii)  $2\sqrt{x+1} - 2\sqrt{2x-5} = \sqrt{x-2}$
7. Sketch the curve  
 $y = \frac{(x+1)(x-2)}{x(x+3)}$